

Neural Information Retrieval on the Official Khmer Legal Corpus

A Comparative Study: From-Scratch BiLSTM vs Frozen Linear Probe vs XLM-R vs PrahokBART Fine-Tuning

Core Research Question & Verdict: When retrieving official statutory articles for conversational citizen queries, does multilingual transfer learning outperform from-scratch and monolingual models?

Key Finding: Full fine-tuning of XLM-R (A3) is the decisive champion (**0.4827 MRR@10**, **51.42% Recall@5**, +44.1% over BM25, $p = 0.0001$). A frozen linear probe (A2) matches BM25 in 5.9s on CPU. Native Khmer PrahokBART (A4) excels at formal titles (80.4% Hit@5) but drops on conversational queries.

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EVALUATION CORPUS

Cambodian Civil Code (2007) & Criminal Code (2009)

SCALE & BENCHMARKS

1,976 Articles | 200 Human Questions | 148 Held-Out Titles

The Citizen Accessibility Bottleneck

- **Domain Problem:** Cambodian citizens struggle to locate relevant statutory protections because laws are lengthy, continuous, and highly technical.
- **Task Definition:** Given an informal Khmer question, retrieve the exact statutory article ($k \leq 5$) across the full 1,976-article corpus without statute filtering.
- **Worked Query Example:**

Query: "តើកិច្ចសន្យាបង្កើតឡើងដោយរបៀបណា?" (How is a contract formed?)

Target: Civil Code Art. 315 ("ការបង្កើតកិច្ចសន្យា")
- **Why Lexical Search Fails:** BM25 depends on exact term matches. On informal queries, vocabulary mismatch causes **57.5%** of BM25 retrieval failures.

The Limon Legacy Font Challenge

Official PDFs from Open Development Cambodia (ODC) use legacy 8-bit **Limon fonts** (`Limon S1`, `Limon R1`):

- Standard PDF tools extract meaningless Latin ASCII: `maRta 336>-` → visual មាត្រា ៣៣៦.-
- **Finite State Machine (FSM) Syllable Reordering:**

Consonant + Subscript (្រ្ទ្រ្ន្រ្ប) + Pre-Vowel → Consonant + Subscript + Vowel

Reorders left-placed visual vowels (្រ, ្ន, ្ប) into canonical Unicode.
- **Guaranteed Integrity:** 100% consecutive extraction of all **1,976 articles** (Civil: 1–1,304, Criminal: 1–672) with complete structural hierarchy.

Dual Evaluation Benchmarks

- **Total Corpus:** 1,976 statutory articles (Civil Code: 1,304, Criminal Code: 672).
- **Primary Benchmark (T-Q: 200 Questions):**
 - 100 Civil Code + 100 Criminal Code questions.
 - Independently authored by Cambodian legal researchers to capture realistic citizen legal dilemmas.
 - Full corpus ranking: 1 query vs 1,976 candidate articles.
- **Secondary Benchmark (T-T: 148 Titles):**
 - 148 held-out article titles retrieving body passages.
 - Tests in-distribution statutory title-matching performance.

Four Strict Leakage Guards (Course Mandate)

- **1. Question Exclusion:** All 200 benchmark questions and their 261 citing articles are **strictly excluded** from the training split.
- **2. Title Stripping:** Statutory titles are stripped from training passages, preventing models from memorizing trivial lexical headings.
- **3. Fixed Splits:** 1,176 training pairs, 130 validation pairs (seed 42). Models selected on validation MRR@10 only.
- **4. Single Test Evaluation:** Test set evaluated strictly once to prevent hyperparameter overfitting on test metrics.

Baseline

BM25 (Sparse)

Tokenizer: khmer-nltk CRF word segmentation.

Params: 0

Training: 0 s

Fast keyword matching; fails on colloquial synonyms.

Approach A1

BiLSTM (Scratch)

Architecture: Random embeddings + 2-layer BiLSTM + mean pooling.

Params: 1.97M (all trainable)

Training: 2,035 s (CPU)

Tests learning Khmer legal semantics without pre-training.

Approach A2 & A3

XLM-R (E5-Base)

A2 Probe: Frozen backbone + 768→768 linear head (0.59M weights, **5.9 s** training).

A3 Fine-Tune: Full end-to-end backprop across **278M params** (7,237 s CPU / 320 s T4).

Multilingual pre-training prior.

Approach A4

PrahokBART

Architecture: Khmer BART encoder + native SentencePiece (32k vocab).

Params: 35.83M (~8x smaller than XLM-R).

Training: 1,602 s (CPU)

Tests compact Khmer-native pre-training vs multilingual giants.

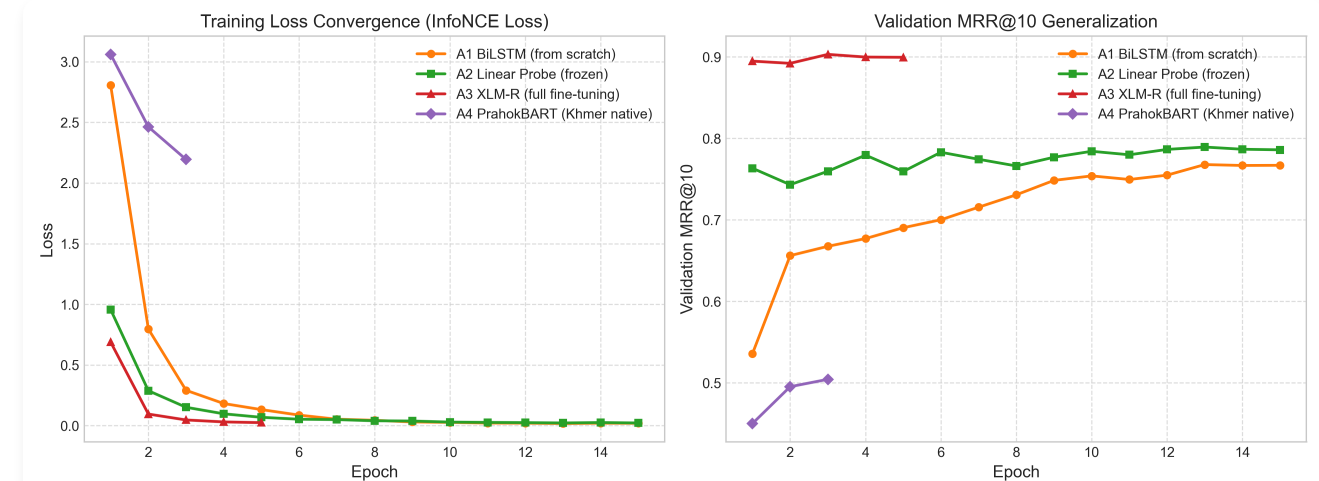
Tokenization Comparison: A1 & BM25 use CRF word segmentation (khmer-nltk, 2,544 vocab), whereas A2 & A3 use 250k multilingual subwords, and A4 uses 32k Khmer SentencePiece subwords.

InfoNCE with In-Batch Negatives

- For mini-batch of B pairs (q_i, p_i) , all non-matching passages p_j ($j \neq i$) serve as negatives:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(q_i, p_i)/\tau)}{\sum_{j=1}^B \exp(\text{sim}(q_i, p_j)/\tau)}$$

- Temperature** $\tau = 0.05$: Fixed sharpening factor pushing representations to compact angular clusters.
- Optimization Protocol:**
 - AdamW (weight decay 0.01) + Linear Warmup (10%) + Cosine Annealing decay.
 - Batch size $B = 16$, fp32 on CPU / fp16 on T4 GPU.



A1 Overfitting: Train loss plunged to 0.017 while val loss stalled at 0.86. From-scratch models cannot learn general semantic manifolds from 1.1k pairs.

A2 vs A3: A2's frozen backbone completely prevents overfitting; A3 fine-tuning achieves smooth convergence with warmup.

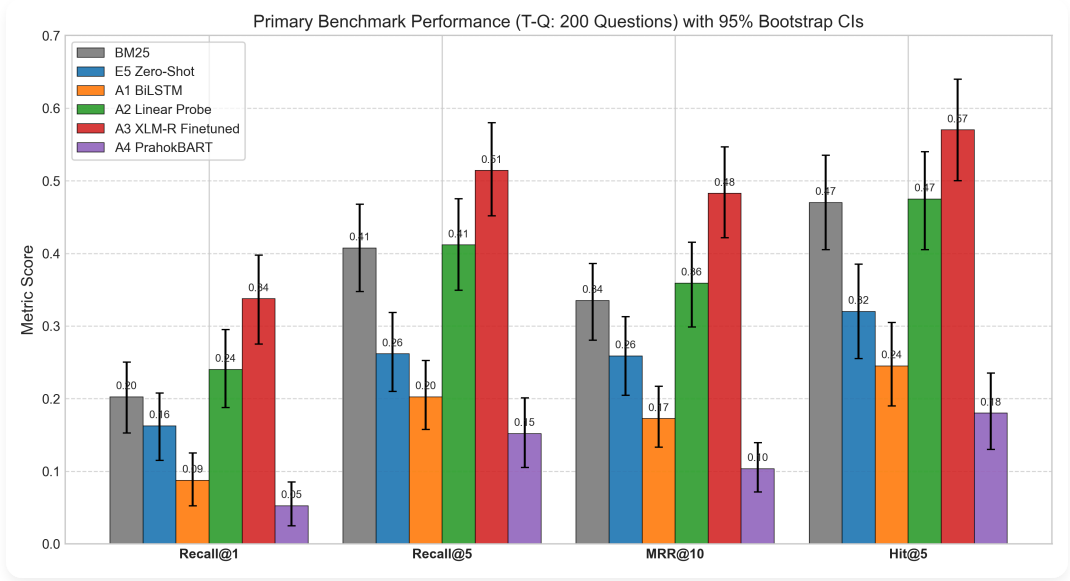
5. Primary Benchmark Leaderboard (T-Q: 200 Questions)

Evaluated on the full 1,976-article corpus with 95% bootstrap confidence intervals ($B = 1,000$):

Rank	Model / Approach	Params	Recall@1 [95% CI]	Recall@5 [95% CI]	MRR@10 [95% CI]	Hit@5
🏆 1	A3: XLM-R Full Fine-Tune	278.04 M	0.3375 [0.28, 0.40]	0.5142 [0.45, 0.58]	0.4827 [0.42, 0.55]	57.0%
🥈 2	A2: XLM-R Linear Probe	0.59 M	0.2400 [0.19, 0.30]	0.4117 [0.35, 0.48]	0.3591 [0.30, 0.42]	47.5%
🥉 3	BM25 Baseline (Lexical)	0	0.2025 [0.15, 0.25]	0.4075 [0.35, 0.47]	0.3350 [0.28, 0.39]	47.0%
4	Multilingual-E5 (Zero-Shot)	278.04 M	0.1625 [0.12, 0.21]	0.2617 [0.21, 0.32]	0.2588 [0.20, 0.31]	32.0%
5	A1: BiLSTM (From Scratch)	1.97 M	0.0875 [0.05, 0.13]	0.2025 [0.16, 0.25]	0.1724 [0.13, 0.22]	24.5%
6	A4: PrahokBART (Khmer Native)	35.83 M	0.0525 [0.03, 0.09]	0.1517 [0.11, 0.20]	0.1035 [0.07, 0.14]	18.0%

Undisputed Winner (A3): Outperforms BM25 by **+44.1% relative MRR@10** and **+86.5%** over zero-shot E5. Dense fine-tuning bridges the colloquial phrasing gap.

Linear Probe Efficiency (A2): Matches BM25 (0.3591 vs 0.3350 MRR) with only 0.59M weights trained in **5.9 seconds**, unlocking low-compute deployment.



Primary metrics across 6 systems with 95% bootstrap error bars.

Paired Bootstrap Hypothesis Tests (10,000 Resamples)

Comparison	Metric	Mean Diff (Δ) [95% CI]	p-value	Sig.
A3 vs BM25	Recall@5	+0.1067 [+0.050, +0.163]	0.0002	***
A3 vs BM25	MRR@10	+0.1477 [+0.096, +0.199]	0.0001	***
A3 vs E5 Zero	MRR@10	+0.2239 [+0.168, +0.280]	0.0001	***
A3 vs A4 (BART)	MRR@10	+0.3792 [+0.320, +0.440]	0.0001	***
A2 vs BM25	Recall@5	+0.0042 [-0.057, +0.063]	0.9031	n.s.

Scientific Proof: Paired testing eliminates inter-query variance. A3 beats BM25 in **72 queries** vs 25 losses ($p = 0.0001$). A2 achieves statistical parity with BM25 ($p = 0.9031$).

The In-Distribution vs Real-World Gap

Model	T-T (148 Titles) Recall@5	T-Q (200 Questions) Recall@5	Generalization Shift
A3: Full Fine-Tune	99.3%	51.4%	-47.9%
BM25 Baseline	97.3%	40.8%	-56.5%
A2: Linear Probe	93.9%	41.2%	-52.7%
A4: PrahokBART	80.4%	15.2%	-65.2%
A1: BiLSTM (Scratch)	75.7%	20.3%	-55.4%
E5 Zero-Shot	72.3%	26.2%	-46.1%

Course Concept: Evaluating only on held-out titles (T-T) creates an illusion of mastery (~99%). Evaluating on conversational queries (T-Q) exposes real distribution shift!

Why Did PrahokBART (A4) Struggle on T-Q?

- **1. Pre-training Task Mismatch:** PrahokBART was pre-trained for sequence-to-sequence text denoising (reconstruction), not sentence-level contrastive embedding.
- **2. Capacity & Cross-Lingual Transfer:** 35.8M parameters / 512-dim vs XLM-R's 278M parameters / 768-dim with massive multi-domain cross-lingual knowledge.
- **3. The T-T Proof:** On formal statutory titles (T-T), PrahokBART hits a strong **80.4% Hit@5** (beating A1 BiLSTM at 75.7% and zero-shot E5 at 72.3%), confirming its Khmer vocabulary quality, but lacks conversational semantic abstraction.

Failure Breakdown at $k = 5$ (200 Questions)

Failure Category	BM25 (106)	A1 (151)	A3 (86)	Impact in A3
Vocabulary Mismatch	61 (57.5%)	79 (52.3%)	52 (60.5%)	-14.8% count
Code Confusion	4 (3.8%)	13 (8.6%)	1 (1.2%)	Nearly zero!
Multi-Article	26 (24.5%)	37 (24.5%)	15 (17.4%)	Spans remedies
Near-Miss (Rank 6-10)	15 (14.2%)	19 (12.6%)	17 (19.8%)	Target in top-10

Code Distinction: A3 made only **1 single Civil/Criminal cross-code confusion** across all 200 questions (0.5% rate vs 8.6% in BiLSTM).

Case 1: Overcoming Vocabulary Mismatch

Query: "តើការចាប់មនុស្សបង្ខំដោយខុសច្បាប់មានទោសអ្វី?" (Unlawful kidnapping)
Target: Criminal Code Art. 253 ("ការចាប់ ការឃុំឃាំង និងការបង្ខំមនុស្សដោយខុសច្បាប់")

- BM25: **Miss (> Rank 50)** — misses colloquial phrasing.
- A3 Fine-Tune: **Rank 1 (Hit)** — maps directly to the kidnapping chapter!

Case 2: Multi-Article Contractual Complexity

Query: "ប្រសិនបើអ្នកលក់មិនព្រមប្រគល់ទំនិញ តើអ្នកទិញអាចទាមទារអ្វីបាន?" (Seller default)
Retrieved by A3: Art. 387 (Demand performance) & Art. 398 (Damages).
Both are legally authentic remedies; highlights need for multi-passage RAG.

Citation-Grounded Khmer Legal RAG Assistant

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[Citizen Query] "តើកិច្ចសន្យាបង្កើតឡើងដូចម្តេច?"
↓ Dense A3 Retriever: < 85 ms (CPU)
↓ Top-3 Articles: Civil Code Art. 315, 316, 317
↓ DeepSeek-V3 LLM (Khmer Prompt): 1.4 s
↓ Citation Guard: Civil Code Art. 315 [VERIFIED]
[Total Latency]: < 1.5 s (UI: FastAPI + Streamlit)
```

Strict citation verification ensures every referenced statute and article number is validated against retrieved passages before display.

Three Core Project Conclusions

- 1. **Pre-Training is Mandatory:** From-scratch BiLSTM completely fails (0.1724 MRR@10) due to extreme vocabulary sparsity in low-resource legal text.
- 2. **Linear Probe is Ultra-Efficient:** A2 matches BM25 (0.3591 MRR) with only 0.59M weights trained in **5.9 seconds** on CPU.
- 3. **Fine-Tuning Delivers Production Quality:** A3 breaks colloquial barriers (**0.4827 MRR@10**, 57% Hit@5, $p = 0.0001$), powering our prototype.

Secondary Benchmark (T-T: 148 Held-Out Titles)

Model	Recall@1	Recall@5	MRR@10	Hit@5
A3: XLM-R Fine-Tune	0.9054	0.9932	0.9456	99.3%
BM25 Baseline	0.7973	0.9730	0.8677	97.3%
A2: XLM-R Linear Probe	0.7905	0.9392	0.8563	93.9%
A4: PrahokBART (Khmer)	0.5946	0.8041	0.6733	80.4%
A1: BiLSTM (Scratch)	0.5135	0.7568	0.6242	75.7%
E5-Base Zero-Shot	0.5405	0.7230	0.6235	72.3%

Computational Footprint & Efficiency

Approach	Total Params	Trainable	Training Time	Hardware
BM25 Baseline	0	0	0 s	CPU
E5 Zero-Shot	278.04 M	0	0 s	CPU
A1 BiLSTM	1.97 M	1.97 M	2,035.3 s	CPU
A2 Linear Probe	278.63 M	0.59 M	5.9 s	CPU
A3 Fine-Tune	278.04 M	278.04 M	7,237.5 s / 320 s	CPU / T4
A4 PrahokBART	35.83 M	35.83 M	1,602.2 s	CPU

Hyperparameter Tuning Grids (Validation MRR@10)

Model	Grid Search Parameters	Best Hyperparameters	Val MRR
A1	Hidden {128, 256} × LR {1e-3, 3e-3} × Drop {0.1, 0.3}	Hidden 128, LR 1e-3, Drop 0.3	0.1706
A2	LR {1e-3, 1e-2} × Weight Decay {0.0, 0.01}	LR 1e-3, Weight Decay 0.01	0.3444
A3	LR {1e-5, 2e-5} × Weight Decay {0.0, 0.01}	LR 2e-5, Weight Decay 0.01	0.4285
A4	LR {5e-5, 1e-4} × Weight Decay {0.01}	LR 5e-5, Weight Decay 0.01	0.2878

Engineering & Q&A Defense Topics

- **Limon FSM Syllable Reordering:** State machine maps 8-bit glyph stream to canonical Unicode consonant-first ordering with zero-gap space handling.
- **InfoNCE In-Batch Negatives:** Contrastive cross-entropy over $B \times B$ similarity matrix avoids expensive negative mining passes during early epochs.
- **100% Deterministic Reproducibility:** Fixed seed 42 enforced across Python, NumPy, PyTorch, and cuDNN; checkpoint save/resume protocol ('last.pt', 'best.pt').